Almost all continued-fraction strings have no nontrivial frequency symmetries

Principia Math In Progress
MathDB 371294

18 September 2026

Abstract

For each fixed length $n \geq 4$, at most $O_n(N^{n-1})$ strings with digits in $\{1, \ldots, N\}$ have a permutation, other than the string and its reverse, with the same Gauss–Kuzmin frequency. This proves Conjectures 4 and 6 of Singh, Zhang, and Hildebrand, including the unordered, distinct-digit formulation in MathDB. The proof reconstructs the order of independent variables, up to reversal, from one homogeneous part of the characteristic polynomial.

Task 1: the full fixed-length theorem

For a string $a = (a_1, \ldots, a_n)$ of positive integers, let $P_{\mathrm{GK}}(a)$ be the Gauss–Kuzmin measure of its continued-fraction cylinder. A symmetry is nontrivial if it gives a permuted string different from both a and its reverse. Let $E_n(N)$ count such strings in $\{1, \ldots, N\}^n$, allowing repeated digits. Let $f(N, n)$ count n-element subsets of $\{1, \ldots, N\}$ having at least one ordering with a nontrivial symmetry, and put $\delta(N, n) = f(N, n)/\binom{N}{n}$ for $N \geq n$.

Theorem 1. *For every fixed integer $n \geq 4$ and every positive integer N,*

$$E_n(N) \leq (n! - 2)(2n - 1)N^{n-1}.$$

Consequently, for $N \geq n$,

$$\delta(N, n) \leq \frac{(n! - 2)(2n - 1)N^{n-1}}{\binom{N}{n}} = O_n(N^{-1}) \longrightarrow 0.$$

The second conclusion is the exact MathDB assertion and [1, Conjecture 6, p. 29]. The first also proves the source's original formulation, Conjecture 4 on p. 7.

1. The known characteristic polynomial

Define the continuant by

$$K() = 1, \qquad K(x_1) = x_1, \qquad K(x_1, \ldots, x_m) = x_m K(x_1, \ldots, x_{m-1}) + K(x_1, \ldots, x_{m-2}) \quad (m \geq 2).$$

For a string of length $n \geq 3$, write

$$Q = K(x_1, \ldots, x_n), \quad U = K(x_1, \ldots, x_{n-1}), \quad V = K(x_2, \ldots, x_n),$$
$$\chi(x_1, \ldots, x_n) = (Q + U)(Q + V).$$

This is the characteristic number of [1, Definition 5.1], regarded as a polynomial in independent variables. Indeed, if $p/q = [0; a_1, \ldots, a_n]$ and p'/q' is the previous convergent, the matrix identity

$$\prod_{j=1}^{n} \begin{pmatrix} 0 & 1 \\ 1 & a_j \end{pmatrix} = \begin{pmatrix} p' & p \\ q' & q \end{pmatrix}$$

gives $q = Q(a)$, $q' = U(a)$, and $p = V(a)$. The cylinder endpoints are p/q and $(p+p')/(q+q')$. Integrating $1/((1+x)\log 2)$ between them gives the established formula [1, Lemma 2.5]

$$P_{\mathrm{GK}}(a) = \left| \log_2 \left(1 + \frac{(-1)^n}{\chi(a)} \right) \right|. \tag{1}$$

Here $\chi(a) > 1$. For each fixed parity the expression on the right is a strictly decreasing function of $\chi(a)$, so

$$P_{\mathrm{GK}}(\sigma(a)) = P_{\mathrm{GK}}(a) \quad \Longleftrightarrow \quad \chi(\sigma(a)) = \chi(a). \tag{2}$$

This equivalence is [1, Lemma 5.2]; it is not a new claim here.

2. Recovering the order from high-degree terms

Let $M = x_1 \cdots x_n$. If $[F]_d$ denotes the homogeneous part of total degree d, the continuant recurrence gives

$$[Q]_n = M, \qquad [Q]_{n-1} = 0, \qquad [Q]_{n-2} = \sum_{i=1}^{n-1} \frac{M}{x_i x_{i+1}}. \tag{3}$$

To see the last formula by induction, the first term of the recurrence extends each adjacent pair in the first $n-1$ positions; the second term supplies the last pair $\{n-1, n\}$. More generally, all degrees occurring in a length-m continuant have the same parity as m. Also $[U]_{n-1} = M/x_n$ and $[V]_{n-1} = M/x_1$. Expand $\chi = Q^2 + Q(U+V) + UV$. The middle term has no part of degree $2n-2$, by the parity observation. Thus

$$[\chi]_{2n-2} = 2 \sum_{i=1}^{n-1} \frac{M^2}{x_i x_{i+1}} + \frac{M^2}{x_1 x_n}. \tag{4}$$

Every displayed quotient is a monomial with nonnegative exponents. For $n \geq 3$, the endpoint pair $\{1, n\}$ is different from all adjacent pairs. Therefore the coefficient of $M^2/(x_i x_j)$, $i < j$, is 2 exactly when the two labels are adjacent in the ordering, 1 for its endpoint pair, and 0 otherwise.

Lemma 1. *For independent variables and $n \geq 3$, the polynomial identity*

$$\chi(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) = \chi(x_1, \ldots, x_n)$$

holds only when σ is the identity or the reversal.

Proof. By (4), polynomial equality forces the two orderings to have exactly the same unordered adjacent pairs. These pairs are the edges of a path on the n distinct variable labels. Such a path has exactly two traversals visiting its vertices in order: start at one endpoint or at the other. The two orderings are consequently equal or reversed. Conversely, reversal preserves χ, either by transposing the displayed matrix product or by continuant reversal symmetry. □

In particular, for every other permutation,

$$F_\sigma(x) = \chi(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) - \chi(x_1, \ldots, x_n)$$

is a nonzero polynomial. Its degree is at most $2n-1$, since the leading term $[\chi]_{2n} = M^2$ cancels.

3. Counting integer zeros

We include the elementary polynomial grid bound to make the counting step self-contained.

Lemma 2. *A nonzero polynomial in m variables over $\mathbb{Q}$ of total degree at most d has at most dN^{m-1} zeros in $\{1,\dots,N\}^m$.*

Proof. For one variable this is the usual root bound. For $m > 1$, write

$$F = \sum_{j=0}^{r} F_j(x_1,\dots,x_{m-1})x_m^j, \qquad F_r \neq 0.$$

The total degree of F_r is at most $d-r$. By induction, F_r vanishes at at most $(d-r)N^{m-2}$ points of the first $m-1$ coordinates. Over these points allow all N values of x_m; over every other point there are at most r roots. The total is at most

$$(d-r)N^{m-1} + rN^{m-1} = dN^{m-1}.$$

Nonzero constants contribute no zeros, as required for the degree-zero case. □

Proof of Theorem 1. If a has a nontrivial symmetry, some index permutation σ other than the identity and reversal satisfies $F_\sigma(a) = 0$, by (2). Lemmas 1 and 2, followed by a union bound over the $n! - 2$ choices, give

$$E_n(N) \le \sum_{\sigma \neq \mathrm{id},\mathrm{rev}} \#\{a \in \{1,\dots,N\}^n : F_\sigma(a) = 0\} \le (n!-2)(2n-1)N^{n-1}.$$

Repeated digits cause no problem: the zero sets may count extra tuples giving trivial string equalities, which only enlarges this upper bound. Each subset counted by $f(N,n)$ has at least one ordered witness counted by $E_n(N)$, and different subsets have disjoint witnesses. Hence $f(N,n) \le E_n(N)$. Finally $\binom{N}{n} \sim N^n/n!$ for fixed n, proving the result. □

Scope. The proof covers every fixed length and the source's intended symmetry condition. It does not establish the stronger length-four prediction $\log f(N,4)/\log N \to 1$ in Conjecture 5(ii) (MathDB 371295). That claim receives no solution credit.

Four proposed follow-up tasks

Task 2: Prove or refute the sharper estimate $f(N,4) = N^{1+o(1)}$. The present general argument gives only $f(N,4) = O(N^3)$.

Task 3: Classify the positive integer solutions with four distinct coordinates of $\chi(\sigma(a)) = \chi(a)$, modulo permutation and reversal, and determine whether finitely many parameterized families account for all such solutions.

Task 4: Determine the sharp growth exponents of $f(N,n)$ for each fixed $n \ge 5$, and identify which permutation equations contribute the largest families of positive integer points.

Task 5: Determine growth rates $n = n(N) \to \infty$ for which $\delta(N, n(N)) \to 0$. Improve the factorial dependence in the union bound by exploiting overlap and symmetry among the equations.

Attribution and validation. The characteristic number, frequency equivalence, and reversal symmetry are credited to [1] and the classical sources cited there. The contribution is the reconstruction in (4) and its use to prove the all-length zero-density assertion. Exact computations check the polynomial identities, finite permutation orbits, and the source's numerical counts. They supplement the general proof rather than prove it by finite enumeration. This manuscript has not received independent mathematical review. The separate audits document the source and bounded prior-claim checks.

References

[1] Shreyas Singh, Zhuo Zhang, and AJ Hildebrand, *An elementary characterization of the Gauss–Kuzmin measure in the theory of continued fractions*, Integers **25** (2025), Article A89, 32 pp., published 17 September 2025. Final published article; DOI 10.5281/zenodo.17144494. See Lemmas 2.5 and 5.2, and Conjectures 4 and 6.